Feasibility Study for

LeitDrive 690V new IGBT module

Evaluation of FF1200R17IP5P with DB1.00 and DB1.51

Outline

1. Motivation
2. Modules and Test Configurations
3. Power Loss Analysis
4. DPT Switching Results
5. Results and Conclusions

Motivation

Original Design Target (Wind)

- $I_{\text{nom}}^{\text{inv}} = 370\,\text{A}$
- $f_{\text{pwm}}^{\text{inv}} = 2.5\,\text{kHz}$
- Overload $\equiv$ wind gust $\Rightarrow I_{\text{max}}^{\text{inv}} \approx I_{\text{nom}}^{\text{inv}}$

New Requirements (Drive Applications)

- $I_{\text{max}}^{\text{inv}} = 1.4 \times I_{\text{nom}}^{\text{inv}} = 518\,\text{A}$
- $f_{\text{pwm}}^{\text{inv}} = 4\,\text{kHz}$ (hard parallelization)

Design Constraint

$$T_{\text{j}}^{\text{design}}(t)\Big|_{\text{max}} \leq T_{\text{j}}^{\text{max}} - 50\,^{\circ}\text{C}$$

Conclusion

Existing modules (DP650B1700, FF650R17IE4) are no longer adequate.

Candidate replacement: FF1200R17IP5P (Infineon, 1200 A / 1700 V, $T_{\text{j}}^{\text{max}} = 175\,^{\circ}\text{C}$).

Modules Under Test

Module	**Manufacturer**	I_{nom}	$T_{\text{j}}^{\text{max}}$	**Status**
DP650B1700	Danfoss	650 A	150 °C	Series production
FF650R17IE4	Infineon	650 A	150 °C	Series production
FF650R17IE4DB2	Infineon	650 A	150 °C	Series production
FF1200R17IP5P	**Infineon**	1200 A	175 °C	**Candidate**

Driver Boards

- **DB1.00** – series production
- **DB1.51** – candidate (new)

Gate Resistances

	R_{g}^{on}	$R_{\text{g}}^{\text{off}}$
DB1.00	1.24 Ω	3.6 Ω
DB1.51	1.175 Ω	1.175 Ω

DPT Configurations

Test	IGBT Module	Driver Board	Type
DPT 1	DP650B1700	DB1.00	Baseline (production)
DPT 2	DP650B1700	DB1.51	New driver board
DPT 3	FF1200R17IP5P	DB1.00	New module
DPT 4	FF1200R17IP5P	DB1.51	**Full candidate config.**

DPT Conditions

$$I_c = 620\,\text{A}, \quad V_{ce} = 630\,\text{V}, \quad T_{case} = 25\,^\circ\text{C}$$

Power Loss Analysis – Operating Conditions

Scenario	$I_{\text{load}}^{\text{inv}}$	$f_{\text{pwm}}^{\text{inv}}$	$f_{\text{load}}^{\text{inv}}$	$T_{\text{hs}}^{\text{inv}}$	$\cos(\varphi)$	**DB**
1	370 A (100%)	4 kHz	15 Hz	65 °C	+0.8 (drive)	DB1.00
2	520 A (140%)	4 kHz	15 Hz	65 °C	+0.8 (drive)	DB1.00
3	370 A (100%)	4 kHz	15 Hz	65 °C	+0.8 (drive)	DB1.51
4	520 A (140%)	4 kHz	15 Hz	65 °C	+0.8 (drive)	DB1.51
5	370 A (100%)	4 kHz	15 Hz	65 °C	−0.8 (gen.)	DB1.00
6	520 A (140%)	4 kHz	15 Hz	65 °C	−0.8 (gen.)	DB1.00
7	370 A (100%)	4 kHz	15 Hz	65 °C	−0.8 (gen.)	DB1.51
8	520 A (140%)	4 kHz	15 Hz	65 °C	−0.8 (gen.)	DB1.51

Junction Temperature – Drive Mode ($\cos\varphi = +0.8$), DB1.00

Scenario 1 – $I_{\text{load}}^{\text{inv}} = 370\,\text{A}$, DB1.00

	T_{j}^{Q}	T_{j}^{D}	P_{c}^{Q}	P_{c}^{D}	P_{sw}^{Q}	P_{sw}^{D}	ΔT_{j}
FF650R17IE4	120 °C	97 °C	242 W	54 W	576 W	175 W	25 °C
FF650R17IE4DB2	130 °C	95 °C	245 W	50 W	598 W	196 W	32 °C
FF1200R17IP5P	110 °C	83 °C	172 W	42 W	705 W	170 W	18 °C

Scenario 2 – $I_{\text{load}}^{\text{inv}} = 520\,\text{A}$, DB1.00

	T_{j}^{Q}	T_{j}^{D}	P_{c}^{Q}	P_{c}^{D}	P_{sw}^{Q}	P_{sw}^{D}	ΔT_{j}
FF650R17IE4	156 °C	108 °C	420 W	85 W	911 W	220 W	42 °C
FF650R17IE4DB2	170 °C	103 °C	430 W	77 W	950 W	236 W	50 °C
FF1200R17IP5P	131 °C	89 °C	280 W	64 W	1000 W	220 W	27 °C

Junction Temperature – Drive Mode ($\cos\varphi = +0.8$), DB1.51

Scenario 3 – $I_{\text{load}}^{\text{inv}} = 370\,\text{A}$, DB1.51

	T_{j}^{Q}	T_{j}^{D}	P_{c}^{Q}	P_{c}^{D}	P_{sw}^{Q}	P_{sw}^{D}	ΔT_{j}
FF650R17IE4	121 °C	98 °C	242 W	54 W	572 W	175 W	25 °C
FF650R17IE4DB2	128 °C	95 °C	245 W	49 W	585 W	198 W	30 °C
FF1200R17IP5P	107 °C	84 °C	172 W	42 W	644 W	171 W	18 °C

Scenario 4 – $I_{\text{load}}^{\text{inv}} = 520\,\text{A}$, DB1.51

	T_{j}^{Q}	T_{j}^{D}	P_{c}^{Q}	P_{c}^{D}	P_{sw}^{Q}	P_{sw}^{D}	ΔT_{j}
FF650R17IE4	156 °C	110 °C	421 W	86 W	905 W	220 W	45 °C
FF650R17IE4DB2	169 °C	103 °C	431 W	77 W	927 W	240 W	50 °C
FF1200R17IP5P	127 °C	90 °C	280 W	65 W	913 W	222 W	25 °C

Junction Temperature – Generator Mode ($\cos\varphi = -0.8$), DB1.00

Scenario 5 – $I_{\text{load}}^{\text{inv}} = 370\,\text{A}$, DB1.00 *(worst case, wind)*

	T_{j}^{Q}	T_{j}^{D}	P_{c}^{Q}	P_{c}^{D}	P_{sw}^{Q}	P_{sw}^{D}	ΔT_{j}
FF650R17IE4	107 °C	124 °C	62 W	210 W	555 W	220 W	30 °C
FF650R17IE4DB2	112 °C	114 °C	62 W	189 W	568 W	215 W	25 °C
FF1200R17IP5P	102 °C	94 °C	45 W	159 W	688 W	179 W	13 °C

Scenario 6 – $I_{\text{load}}^{\text{inv}} = 520\,\text{A}$, DB1.00 *(worst case, wind)*

	T_{j}^{Q}	T_{j}^{D}	P_{c}^{Q}	P_{c}^{D}	P_{sw}^{Q}	P_{sw}^{D}	ΔT_{j}
FF650R17IE4	130 °C	155 °C	100 W	343 W	841 W	283 W	50 °C
FF650R17IE4DB2	137 °C	134 °C	104 W	299 W	862 W	277 W	35 °C
FF1200R17IP5P	118 °C	107 °C	72 W	247 W	964 W	240 W	32 °C

Junction Temperature – Generator Mode ($\cos\varphi = -0.8$), DB1.51

Scenario 7 – $I_{\text{load}}^{\text{inv}} = 370\,\text{A}$, DB1.51 *(worst case, wind)*

	T_{j}^{Q}	T_{j}^{D}	P_{c}^{Q}	P_{c}^{D}	P_{sw}^{Q}	P_{sw}^{D}	ΔT_{j}
FF650R17IE4	106 °C	124 °C	62 W	210 W	548 W	200 W	22.5 °C
FF650R17IE4DB2	111 °C	114 °C	62 W	189 W	557 W	219 W	22.5 °C
FF1200R17IP5P	100 °C	94 °C	45 W	159 W	629 W	181 W	12 °C

Scenario 8 – $I_{\text{load}}^{\text{inv}} = 520\,\text{A}$, DB1.51

	T_{j}^{Q}	T_{j}^{D}	P_{c}^{Q}	P_{c}^{D}	P_{sw}^{Q}	P_{sw}^{D}	ΔT_{j}
FF650R17IE4	129 °C	155 °C	103 W	343 W	836 W	283 W	50 °C
FF650R17IE4DB2	136 °C	135 °C	104 W	299 W	843 W	282 W	35 °C
FF1200R17IP5P	114 °C	107 °C	72 W	247 W	878 W	243 W	30 °C

Switching Energy – Reference vs. Measured

Reference Values (from datasheet, scaled to test conditions)

	T_{case}	E_{on}^{ref}	E_{off}^{ref}	I_c	V_{ce}
FF650R17IE4	25 °C	136 mJ	93 mJ	620 A	630 V
FF1200R17IP5P	25 °C	132 mJ	100 mJ	620 A	630 V

Measured – DB1.00

	E_{on}^{m}	E_{off}^{m}
FF650R17IE4	80 mJ	90 mJ
FF1200R17IP5P	105 mJ	152 mJ

Measured – DB1.51

	E_{on}^{m}	E_{off}^{m}
FF650R17IE4	91 mJ	93 mJ
FF1200R17IP5P	107 mJ	130 mJ

DPT Summary – Switching Parameters

	Case 1	Case 2	Case 3	Case 4
	DP650 + DB1.00	DP650 + DB1.51	FF1200 + DB1.00	FF1200 + DB1.51
E_{or}	80	91	105	107
E_{off} [mJ]	90	93	152	**130**
I_{rr} |	512	**448**	488	**464**
Δu_{ce} [V]	245	187	173	**86**

FF1200 + DB1.00 (Case 3)

Higher E_{off} (152 mJ) compared to reference. Asymmetric R_g may be sub-optimal.

FF1200 + DB1.51 (Case 4)

Best Δu_{ce} (86 V). Reduced E_{off} vs Case 3. Lowest voltage overshoot overall.

Results Overview – FF1200R17IP5P vs. Existing Modules

Metric	FF650R17IE4	FF650R17IE4DB2	FF1200R17IP5P
T_j^{max} allowed	150 °C	150 °C	175 °C
T_j^{Q} @ 520 A, drive	156 °C	170 °C	131 °C
T_j^{D} @ 520 A, gen.	155 °C	134 °C	107 °C
ΔT_j @ 520 A, drive	42 °C	50 °C	27 °C
Design rule satisfied (140% load)	**No**	**No**	**Yes**

Conclusions

FF1200R17IP5P + DB1.51 – Recommended

- Satisfies the thermal design rule at 140% load
- Lower T_j than existing modules in all scenarios
- Lowest $\Delta u_{ce} = 86\,\mathrm{V}$ (Case 4)
- Compatible with DB1.51 driver board
- $T_j^{max} = 175\,^\circ\mathrm{C}$ provides larger thermal margin

Existing Modules – Inadequate

- FF650R17IE4 and FF650R17IE4DB2 exceed the thermal limit at 140% load
- $T_j^{max} = 150\,^\circ\mathrm{C}$ insufficient for new requirements

Next Steps

- Load testing under real operating conditions
- Thermal cycling validation
- Platform integration testing

Thank you for your attention

Questions?

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690 V LeitDrive Platform – IGBT Module Feasibility Study

FF1200R17IP5P (Infineon) – DPT Evaluation with DB1.00 and DB1.51